

Two Attacks on the Dead-Nucleus Detection Deficit: Detection-First Architectures Do Not Transfer, and 2× Resolution Buys Dead at a Priced Trade

Stage Report II (Course Project bdd26)

October 1, 2026

Abstract

Stage report I located the PanNuke *Dead* (apoptotic) class deficit in *detection* —38.5% of Dead ground-truth nuclei receive no overlapping prediction—and showed it survives every loss-level and data-level intervention we tried, across three dense-decoder architectures. Two escapes were left untested: the *detection-first* architecture family, whose published Dead numbers are the strongest in the literature, and *working resolution*. This report tests both under the same strict protocol (official 3-fold split, last checkpoint, validation-only tuning, artifact-aware Dead reporting). **Line A (negative)**. The released per-fold KongNet checkpoints—a six-head CenterNet-style detector whose fold mapping we first resolve empirically—score 0.346 mPQ under a validation-tuned decode, 11 points below the *weakest* dense decoder; its contour-cut channel is net-harmful at 40× (every optimum disables it), same-class touching nuclei fuse (merged rate 0.12–0.18 vs 0.04–0.05 for watershed decoders), interior-Dead miss is *worse* (0.50 vs 0.35), and the paper’s Dead F_c 0.59 falls to 0.34. LKCell-L, the strongest released per-fold weights we could find, reproduces uniformly ~ 0.015 below its paper and, under matched conditions, is a *better segmenter but worse typer* than our CellViT-UNI baseline (bPQ +0.008, Dead PQ −0.022): architecture strength does not convert into Dead or typing gains. **Line B (a priced trade)**. Doubling working resolution to 512px as a single-variable change makes every pre-registered Dead endpoint positive (interior-Dead miss 0.35→0.25) at a uniform bPQ tax of −0.026. Error analysis refutes the downsample hypothesis and traces most of the tax to a hidden confound: the watershed decode applies *fixed pixel-unit constants* (minimum object size, Sobel aperture) that silently relax in native units at 2×. The pre-registered fix—scaling those constants by the resolution factor—recovers 50–87% of the mPQ/bPQ/mPQ⁺ cost while keeping (slightly enlarging) the Dead gain; a validation-only post-processing lever (same-class instance merging + edge-sliver dropping) then passes a strict test improvement on all 9 runs of a 3×3 split×seed grid. The pre-registered seed-noise verdict on that grid: the *prices* are seed-robust (4.7–5.3× the largest seed std, same sign on every run), the *Dead gain* is fragile (0.4–0.7× the per-seed Dead std; one split’s seed-mean flips sign). Final standing: −0.008 mPQ / −0.005 bPQ buys −0.069 interior-Dead miss, +0.034 Dead F_c , and mPQ⁺ parity—not a default win, an explicitly priced trade. We conclude that the Dead deficit is not an architecture-family artifact but a resolution×separation problem that current decoders price in different currencies, and that pixel-unit decode constants are a first-class confound that any resolution study must control for.

1 Introduction

Stage report I ended with a closed diagnosis: on PanNuke [1], the Dead (apoptotic-body) class fails at *detection*, not classification. The foreground head is silent on 83% of missed Dead nuclei while the type branch already assigns them $P(\text{Dead}) \approx 0.26$; re-typing with a pathology VLM, training-free recovery, loss re-weighting, copy-paste augmentation, and in-context synthesis were all null or harmful; and the deficit was shared by all three dense-decoder architectures we evaluated (a ViT-L/16, a full-resolution ResNet-50, a ConvNeXt decoder). Two hypotheses remained open, and this report closes both under the same pre-registered, noise-calibrated protocol:

1. **Architecture (Line A):** the failure is a property of *dense decoders* (foreground + watershed). Detection-first designs that predict per-class centroids and assemble masks around them report by far the strongest Dead numbers in the literature (KongNet [7]: per-nucleus Dead F_c 0.59, vs 0.23–0.36 for our dense models under strict decoding). If that advantage is real under matched evaluation, the Dead deficit has an architectural fix. We test the released per-fold KongNet checkpoints end-to-end—resolving their undocumented fold mapping empirically first—and re-evaluate LKCell-L [8], the strongest released per-fold weights of the CellViT lineage, in the same pipeline.
2. **Resolution (Line B):** the failure reflects the 256 px working resolution: Dead nuclei are small (median 107 px vs 368 for other classes) and 31% of interior Dead nuclei are missed even at ≥ 256 px, so more pixels should help detection regardless of decoder. We double the working resolution of the unchanged CellViT-UNI recipe (single-variable change), then follow the error trail wherever it leads: an initial uniform bPQ tax, a mechanism diagnosis that overturns the obvious explanation, a pre-registered decode fix, a validation-only post-processing lever, and a 3×3 split $\times$ seed grid whose seed-noise verdict was written down before the last runs landed.

Contributions:

1. **A strict-protocol verdict on detection-first architectures** (Sec. 4): KongNet’s released per-fold weights, given their best validation-tuned decode, are 11 mPQ points below the weakest dense decoder, fuse touching nuclei, and miss *more* interior Dead nuclei; the paper’s Dead F_c 0.59 does not replicate (0.34 strict). LKCell-L is a better segmenter and worse typer under matched conditions. Generalisable lesson: published Dead advantages so far reflect evaluation recipes and segmentation–typing trade-offs, not a solved Dead detector.
2. **A mechanism study of resolution in nuclei decoding** (Sec. 5.2–5.3): the $2 \times$ bPQ tax is size-axis-monotone, not class-localised; boundary quality of matched nuclei is unchanged; and the dominant cause is two pixel-unit constants hardcoded in the watershed decode (object-size floor, Sobel aperture) that relax in native units when resolution doubles. Scaling them by the resolution factor ($du2$) recovers 50–87% of the aggregate cost. To our knowledge this confound is undocumented in the nuclei-segmentation literature, and it applies to *any* pipeline built on the HoVer-Net post-processing stack.
3. **A pre-registered, fully replicated priced-trade characterisation of $2 \times$ resolution** (Sec. 5.4–5.6): a validation-only lever (same-class merge + edge-sliver drop, frozen constants) strictly improves the $du2$ arm on all 9 grid runs; the 3×3 seed grid separates robust prices from a fragile Dead gain; and the final standing prices every Dead improvement in mPQ/bPQ units.

4. **Two further evaluation-methodology results:** the KongNet fold mapping of released checkpoints can be recovered from memorisation alone (train-fold detection F1 margin ≥ 0.13 , no test-fold information used); and LKCell-L’s paper-to-strict gap is *uniform* (~ 0.015 on every headline metric), the signature of a protocol translation rather than a pipeline error—the second released model, after HoVer-NeXt-T in report I, whose recipe gap exceeds typical leaderboard gaps.

Everything runs in one codebase; every number comes from the same bit-exact evaluator, and all tables in this report were re-derived from stored prediction artifacts on 2026-10-01 (script `scripts/report2_numbers.py`, output archived in the repository).

2 Related Work

Dense-decoder lineage. HoVer-Net [2] established the nuclear-probability / horizontal-vertical / type decomposition with watershed decoding that CellViT [3] inherited; HoVer-NeXt [5] and LKCell [8] are recent variants. Report I showed this whole family shares the Dead detection deficit.

Detection-first and point-prediction models. Centroid/heatmap detectors for nuclei go back to StarDist-class and CenterNet-style designs; KongNet [7] (a six-head EfficientNetV2-L model: per-class heatmap + regression + segmentation/contour channels) reports the strongest published per-nucleus Dead F_c (0.59) with released per-fold weights, which is what made it the right Line-A test: the claim is specific, checkable, and load-bearing for the “detection-first fixes Dead” hypothesis. PromptNucSeg [6] and CellVTA represent the prompt/adaptor branch; their released weights are queued for the same re-evaluation (Sec. 7).

Resolution in nuclei segmentation. Higher input or working resolution is an evergreen lever (CellViT-SAM-H at 512; LKCell’s multi-scale large kernels; HoVer-NeXt’s pyramid), but published ablations compare end-to-end configurations, not the resolution variable with the decode held accountable. The closest antecedent to our finding is the classic scale-space literature: watershed markers and morphological floors are pixel-unit operators, so their effective scale changes with image resolution—a fact that, to our knowledge, has not been called out as a confound in learned nuclei-segmentation pipelines, where these constants are inherited verbatim from HoVer-Net’s released post-processing code.

3 Experimental Setup

Protocol (unchanged from report I). PanNuke folds 1–3, official assignment (train/val/test = 1/2/3, 2/1/3, 3/2/1); last checkpoint for our trained runs, released per-fold (validation-selected) checkpoints for the re-evaluated models; no test-fold tuning of anything; official tissue-averaged mPQ/bPQ from our bit-exact evaluator; TTA (8 dihedral views, HV remapped) reported separately. Dead endpoints are reported four ways as in report I: official Dead PQ, Uterus-excluded Dead PQ (the official per-class PQ averages only over images containing the class, and fold 3 contains 34 Uterus images near zero for every model), per-nucleus Dead F_c (12px centroid matching), and the interior (non-border) Dead miss rate.

Released-weight re-evaluation rules (Line A). (1) The fold mapping of released checkpoints must be established without touching test folds: for KongNet we run each checkpoint on 200-image subsamples of every *train* fold and read the mapping off the memorisation diagonal (Sec. 4.1). (2) Any decode thresholds are tuned on validation folds only, with grids extended until the optimum is bracketed; test folds run once with frozen constants. (3) Class-channel semantics of vendored decoders are verified on train folds only (matched-pair confusion diagonal ≥ 0.97). LKCell-L uses its released configuration (normalisation, 256 px decode) unmodified through our shared pipeline.

2× working resolution (Line B). The exact CellViT-UNI recipe with the working resolution doubled to 512: bilinear image upsample after augmentation, nearest-neighbour labels, prediction post-processed at 512 and downsampled to 256 for the standard evaluator (instance type-probability tables computed at model resolution). One variable changes. *du2* denotes the decode fix of Sec. 5.3; *lever* the post-processing of Sec. 5.4. The seed grid is 3 splits $\times$ seeds {19, 1, 2}; baseline seeds 1/2 exist on split 1 (report I), so seed standard deviations are fully populated there and reported per split.

Noise calibration (carried over, now twice validated). Seed std of the identical CellViT-UNI recipe: 0.0021 mPQ but 0.0074 Dead PQ (split 1); image-bootstrap CI half-width on Dead 0.02–0.03. Verdict rules: Dead changes below ~ 0.01 –0.015 single-seed are noise; every intervention verdict is a multi-run statement with paired, tissue-stratified image bootstrap 95% CIs; validation-fold decisions only. The Line-B seed-noise verdict was pre-registered before the last grid runs landed (Sec. 5.5).

4 Line A: Detection-First Architectures Under the Strict Protocol

4.1 KongNet: fold mapping, best decode, and a decisive negative

Assets and fold mapping. KongNet’s released `KongNet.PanNuke_{1,2,3}.pth` (176.9M parameters, `tf_efficientnetv2_L` encoder + six SMP decoders: one overall head + five per-class heads, each with heatmap/segmentation/contour channels) ship with no fold documentation. Running all three checkpoints on 200-image subsamples of every *train* fold gives a clean memorisation diagonal (pooled detection F1, centroid pairing at 12 px): checkpoint *k* wins train fold *k* at 0.905/0.944/0.912 with margin ≥ 0.13 over both held-out folds—`_k.pth` is official split *k*, resolved without any test-fold information. Class-channel semantics were verified on train folds (matched-pair class agreement 0.97–0.99, Dead diagonal ≈ 0 off-diagonal).

The contour cut is net-harmful at 40×. The released inference recipe assembles instances as per-class segmentation-minus-contour masks. We swept the two decode thresholds on validation folds (grids extended twice until each optimum was bracketed). Every checkpoint’s optimum *disables* the contour cut entirely (contour threshold 1.0): on checkpoint 1, disabling the cut moves validation mPQ 0.331 $\rightarrow$ 0.345 and bPQ 0.457 $\rightarrow$ 0.473. The repo’s contour-threshold-0.3 recipe over-splits nuclei at this scale; with the cut off, the decode degenerates to plain per-class connected components.

Test verdict. Table 1, top block: with the frozen best decode, KongNet reaches **0.346 mPQ / 0.471 bPQ**—11 and 17–19 points below the weakest dense decoder—and every pre-registered

Table 1: Strict-protocol architecture table (3-split means, no TTA, last/released per-fold checkpoints, validation-tuned decodes). Dead endpoints: official PQ, Uterus-excluded PQ, per-nucleus F_c , DQ^+ , and interior-Dead miss (share of non-border Dead GT with no matched prediction; mean of per-split rates). HoVer-NeXt-T (report I: 0.458 mPQ / 0.637 bPQ / 0.136 Dead PQ) lacks artifact-derived Dead columns here. LKCell-L paper row as published.

Model	mPQ	bPQ	mPQ ⁺	Dead PQ	Dead −Uterus	Dead F_c	Dead DQ^+	int-Dead miss
KongNet (strict)	.3460	.4706	.3472	.053	.070	.344	.232	.520
HoVer-Net (ours)	.4564	.6613	.4430	.127	.152	.230	.275	.404
LKCell-L (strict)	.4923	.6729	.5235	.154	.201	.382	.418	.347
paper	.508	.685	—	.172	—	—	—	—
CellViT-UNI (ours)	.4995	.6653	.5178	.176	.218	.359	.396	.372
x2 + du2	.4894	.6583	.5157	.183	.226	.395	.424	.302
x2 + du2 + lever	.4910	.6608	.5171	.183	.227	.393	.422	.302

Dead endpoint fails: official Dead PQ 0.053 (dense 0.127–0.176), Uterus-excluded 0.070, interior-Dead miss 0.50 *vs* 0.354 (CellViT-UNI) and 0.413 (HoVer-Net) on the shared splits-1+3 frame. Only per-nucleus Dead F_c is mid-pack (0.344). The error profile explains why: *same-class touching nuclei fuse*. The merged-GT rate is 0.121–0.175 across splits (CellViT-UNI 0.044–0.045, HoVer-Net 0.048–0.050); with the contour cut disabled the decode cannot separate touching instances at all, and re-enabling it costs more than it saves. This is precisely the failure geometry of karyorrhectic Dead clusters (report I: 73–82% of merged Dead touch another Dead): the Dead deficit is an *instance-separation* problem, not a detection-density problem, so a detector that finds Dead nuclei but cannot cut fragment clusters converts nothing into Dead PQ. KongNet’s paper Dead F_c 0.59 (their decode, their evaluation) does not survive validation-tuned decoding; Line A is closed as a negative result.

4.2 LKCell-L: a uniform paper gap, and segmentation–typing non-conversion

LKCell-L (UniRepLKNet-S encoder, RepLK decoder, CellViT np/hv/tp heads) is the strongest model with released per-fold weights in the CellViT lineage (paper: 0.508 mPQ / 0.685 bPQ / 0.172 Dead PQ). We ran its released per-fold `model_best` checkpoints through the shared pipeline (its normalisation, its 256px decode, no TTA):

- **The paper gap is uniform:** strict 0.4923/0.6729/0.154 *vs* published 0.508/0.685/0.172— $-0.016/-0.012/-0.018$ on every headline metric. A fold or normalisation error craters metrics; a uniform shift is the signature of a protocol translation (checkpoint selection, decode, TTA recipe). Recorded as-is; it cannot be resolved further from the paper alone.
- **Under matched conditions, no Dead gain:** *vs* our CellViT-UNI baseline (Table 1), LKCell-L is -0.007 mPQ and -0.022 Dead PQ while $+0.008$ bPQ and $+0.006$ mPQ⁺. Decomposition: it *finds* Dead nuclei better (Dead F_c 0.382 *vs* 0.359; interior-Dead miss 0.347 *vs* 0.372) but its detected Dead nuclei still fail type pairing—better detection does not become better Dead PQ. Same direction as KongNet: general architecture strength does not transfer into Dead or typing gains under a matched protocol.

Notably, our costliest arm (x2+du2+lever) sits at mPQ parity with strict LKCell-L (0.4910 *vs* 0.4923) while winning every Dead endpoint (interior-Dead miss 0.302 *vs* 0.347; Dead F_c 0.393 *vs*

Table 2: B1 initial result (split 1, TTA): $2\times$ resolution vs the 3-seed baseline (seed std in parentheses; Δ with paired, tissue-stratified image-bootstrap 95% CIs).

Metric	base (3 seeds)	x2 (seed 19)	Δ [95% CI]
mPQ	.4979 (.0032)	.4800	−.0180 [−.0238, −.0122]
bPQ	.6708 (.0018)	.6447	−.0261 [−.0308, −.0219]
Dead PQ	.1429 (.0054)	.1614	+.0185 [+.0019, +.0364]
Neoplastic PQ			−.0358
Epithelial PQ			−.0208
Connective PQ			−.0201
Inflammatory PQ			−.0109
Pre-registered Dead endpoints (TTA):			
Uterus-excluded Dead PQ	.198	.219	+.021
interior-Dead missed-bg	.352	.251	−.101 ($\sim 10\times$ seed spread)
interior-Dead matched	.50	.56	+.06
Dead F_c	.310	.356	+.046
Dead DQ ⁺	.352	.371	+.019
Uterus Dead PQ (34 img)	.041	.055	+.014

0.382; Dead DQ⁺ 0.422 vs 0.418).

5 Line B: $2\times$ Working Resolution

5.1 B1: every Dead endpoint positive, a uniform bPQ tax

Table 2: the resolution hypothesis passes on the Dead side—every pre-registered Dead endpoint is positive, and the interior-Dead miss drop ($0.352 \rightarrow 0.251$) is an order of magnitude beyond seed spread, landing exactly where report I’s artifact analysis placed the class-specific deficit. But the pre-registered bPQ non-regression fails decisively: $-0.026 = 13\times$ the seed std, negative in all 19 tissues, and *every* non-Dead class regresses. Under the pre-registered combined endpoint, x2-as-trained is not a win; the rest of Line B is the anatomy of that tax.

5.2 Mechanism: a size axis and two hidden constants

Three observations locate the mechanism (Figs. 1–2):

1. **The tax is not colocated with the Dead gain.** Across the 97 Dead-bearing test images, $\text{corr}(\Delta\text{Dead}, \Delta\text{bPQ}) = +.23$; images that gained Dead are bPQ-neutral (median $\Delta\text{bPQ} +.003$), images without Dead gains carry the loss (median $-.027$). This already rules out “Dead improvements cannibalise other classes”.
2. **Size, not class, is the axis.** Match rate moves $+.104$ for non-Dead nuclei under 200 px (Dead 0–200: $+.091$) and declines monotonically with size to $-.115$ at 3200–6400 px. Boundary quality of *matched* nuclei is unchanged (matched-pair IoU $\pm .01$ in every bin; missed-poor-shape rates do not grow), so nuclei are not being matched worse—large ones are being *cut into pieces*. Error composition confirms it: GT split-rate roughly doubles in every class (Neoplastic $.009 \rightarrow .017$, Connective $.013 \rightarrow .025$, Dead $.010 \rightarrow .026$); among large (≥ 1600 px) non-Dead nuclei split goes $.072 \rightarrow .104$, merged $.003 \rightarrow .009$, missed-bg $.024 \rightarrow .054$; predicted-instance count rises $25.5 \rightarrow 28.4$ per image and the under-80 px share of predictions triples

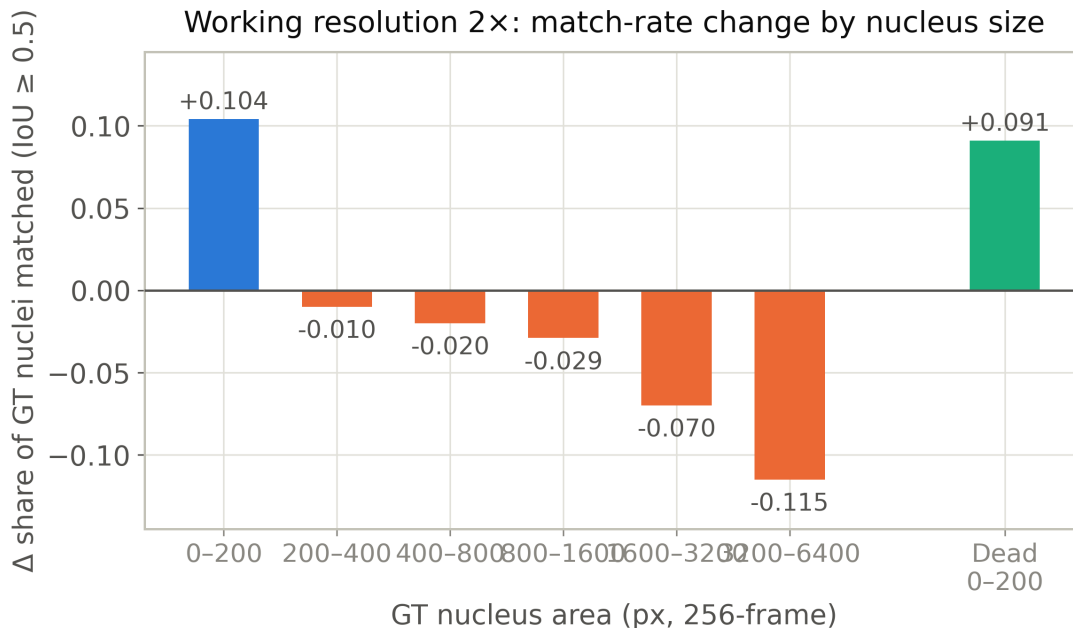


Figure 1: Match-rate change from 2 $\times$ resolution by GT nucleus area (split-1 TTA, x2 seed 19 minus 3-seed baseline mean; IoU ≥ 0.5). The effect is a monotone function of *size*, not class: every nucleus below 200px gains (Dead included, +.091), every size band above 400px loses, monotonically. Matched-pair IoU is flat (± 0.01) in every bin.

(.034 $\rightarrow$.109); prediction-side fp-split roughly doubles (.019 $\rightarrow$.037 pooled). Visual review of montages: 11/12 sampled damaged large nuclei are interior-fragmentation mosaics (one nucleus $\rightarrow$ up to 7 pieces) with the outer contour still on the ground truth, while 12/12 sampled small-Dead gains are genuine detections (mean IoU 0.70).

3. **The obvious explanation is wrong.** "The 512 $\rightarrow$ 256 nearest-downsample of the instance map destroys boundaries" is refuted by the matched-IoU parity in every size bin. The actual cause sits in the decode: HoVer-Net's post-processing applies *fixed pixel-unit constants* to the predicted maps—`remove_small_objects(min.size=10)` (area units; 2.5 native px² at 2 $\times$) on the blob and marker maps, and a Sobel operator with aperture $k = 21$ (10.5 native px) on the HV maps. At 2 $\times$ every one of these thresholds silently *relaxes* in native units: spurious markers survive on chromatin texture and fragment large nuclei, and tiny blobs pass the size floor into the prediction set. The small-nucleus detection gain, by contrast, is a genuine model-resolution benefit.

5.3 The pre-registered decode fix (*du2*)

The fix follows from the diagnosis and was registered before running it: scale the pixel-unit constants by the resolution factor u —`min.size` $\rightarrow 10u^2$, Sobel aperture $\rightarrow$ the nearest odd multiple of $21u$ (41). (OpenCV caps Sobel apertures at 31; beyond that we linearly resample the official 21-tap operator's impulse-response kernels, exact at $u=1$, unit-tested bit-exact.) Re-decoding the existing checkpoints—no retraining:

Table 3/Fig. 3: the two constants own most of the tax—50–87% of the mPQ/bPQ/mPQ⁺ cost depending on metric and TTA—and the Dead gain is kept and slightly enlarged (no-TTA

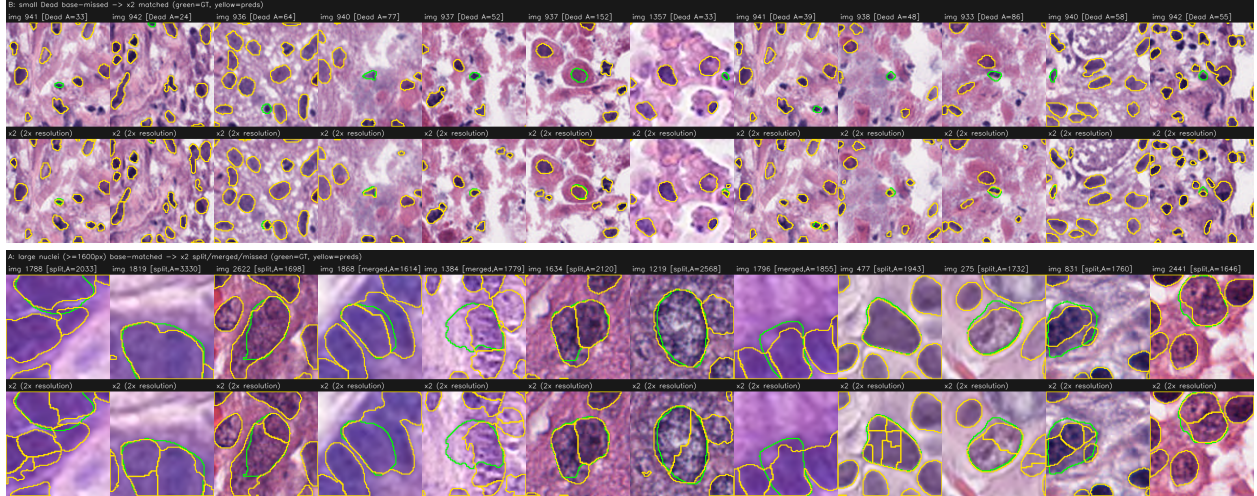


Figure 2: Top (gains): Dead/small nuclei missed at 256 (top row of each pair: no prediction outline) are detected at 512 (bottom row); green = ground truth, yellow = predicted instances. Bottom (damage): large nuclei matched at 256 (single outline) fragment at 512 into interior mosaics—multiple yellow rings inside one green ring—while the outer boundary still tracks the ground truth.

Table 3: *du2* recovery (3-split means, seed 19, last checkpoint). % = share of the x2-vs-base gap recovered by the decode fix alone. Dead keeps its gain (and slightly grows).

	Arm	mPQ	bPQ	mPQ ⁺	Dead PQ
no TTA	base 256	.4995	.6653	.5178	.1760
	x2	.4791	.6413	.5012	.1816
	x2 + du2	.4894	.6583	.5157	.1831
	recovered	50%	71%	87%	(gain grows)
TTA	base 256	.5047	.6706	.5227	.1767
	x2	.4848	.6488	.5070	.1786
	x2 + du2	.4926	.6620	.5189	.1810
	recovered	39%	61%	76%	(gain grows)

+0.006→ +.007). Exactly as predicted from the mechanism (boundary quality was already at parity; the Dead gain lives in detection, not decode). Per split, the x2 Dead gain is fold-3-driven (split-1 +.017, split-2 +.009 no TTA) while split-3 (test fold 1) shows a small loss (−.009) that *du2* largely recovers (−.003)—consistent with the known fold-composition asymmetry of Dead. The residual gap (−.007 bPQ no TTA) is genuine fp-split at 512, not decode: scaling the third constant (the marker-open kernel, *du2mk2*) was evaluated on validation and *rejected* (worse mPQ at equal bPQ).

5.4 A validation-only lever: strict improvement on all nine runs

The visual analysis of the residual suggested two decode-side moves: merge same-class adjacent instances (the mosaics have clean outer boundaries) and drop edge-touching slivers (a third of the tiny FPs are border-clipping fragments). Both have knobs; both were selected on *validation* folds only, on the *du2* arm, under the rule “bPQ ≥ baseline − .002, then max mPQ”. All three splits’

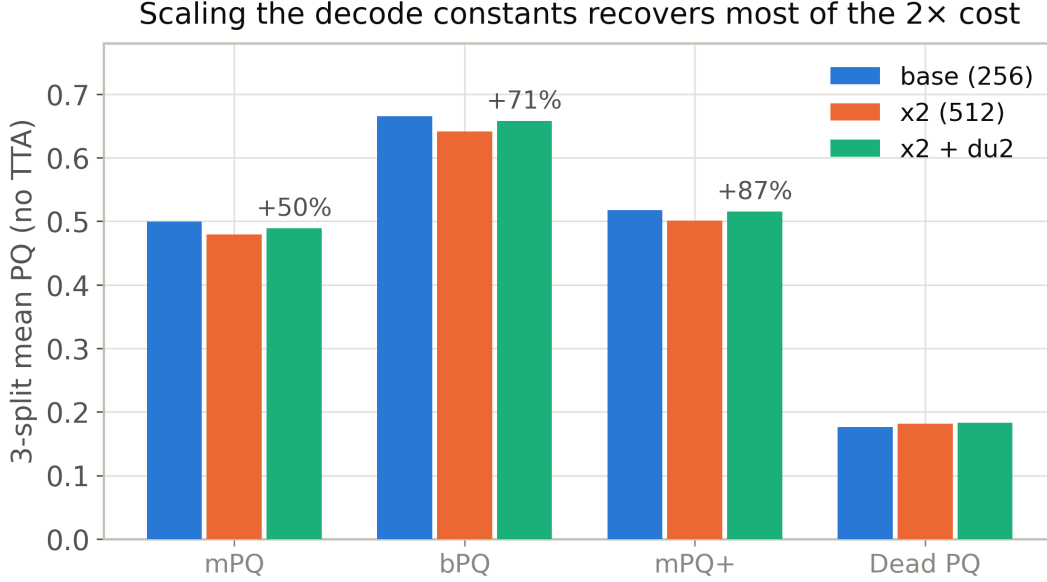


Figure 3: Table 3 (no TTA): scaling two decode constants recovers most of the aggregate 2× cost while the Dead gain survives. mPQ⁺ closes to within .002 of the baseline.

Table 4: Lever verdict ($du2lev - du2$, paired image-bootstrap 95% CIs). Top: the three original splits (seed 19). Replication: positive mPQ and bPQ deltas on **9/9** grid runs (Sec. 5.5); Dead unchanged everywhere (CIs straddle 0).

Split	ΔmPQ [CI]	ΔbPQ [CI]	$\Delta Dead PQ$ [CI]
1	+0.0020 [+0.0014, +0.0026]	+0.0030 [+0.0022, +0.0041]	+0.0015 [−.0008, +0.0051]
2	+0.0016 [+0.0009, +0.0022]	+0.0025 [+0.0018, +0.0032]	+0.0005 [−.0012, +0.0028]
3	+0.0014 [+0.0008, +0.0021]	+0.0021 [+0.0014, +0.0029]	−.0018 [−.0055, +0.0016]

validation sweeps—and a fourth, extended sweep on a decode variant—independently choose the same pair: merge fraction 0.25, sliver area floor 40 px.

Frozen, applied to test:

Table 4: a strict test improvement—all six mPQ/bPQ CIs exclude zero on the positive side, every Dead view within ± 0.002 , predicted instances drop 27.2→26.6 per image (the lever removes ~ 0.5 fragments and slivers per image and nothing else measurable). Selection generalised with zero test-fold tuning.

5.5 The 3×3 seed grid and the pre-registered seed-noise verdict

The grid (Table 5, Fig. 4) was completed for the pre-registered seed-noise verdict: compare each Line-B effect against the per-seed standard deviation of the identical recipe.

- **The prices are seed-robust.** $\Delta mPQ -0.0098$ and $\Delta bPQ -0.0069$ ($du2$ vs base, 3-split means) are $4.7\times$ and $5.3\times$ the largest per-split seed std (0.0021 / 0.0013), same sign on every one of the 9 runs. x1 stays ahead on mPQ/bPQ at $\sim 5\times$ seed noise.
- **The Dead gain is fragile.** +0.0049 on 3-split means is only $0.4\text{--}0.7\times$ the per-seed Dead std (0.0068–0.0112). Paired same-seed comparisons favour $du2$ on 4 of 5 pairs (split-3 seed

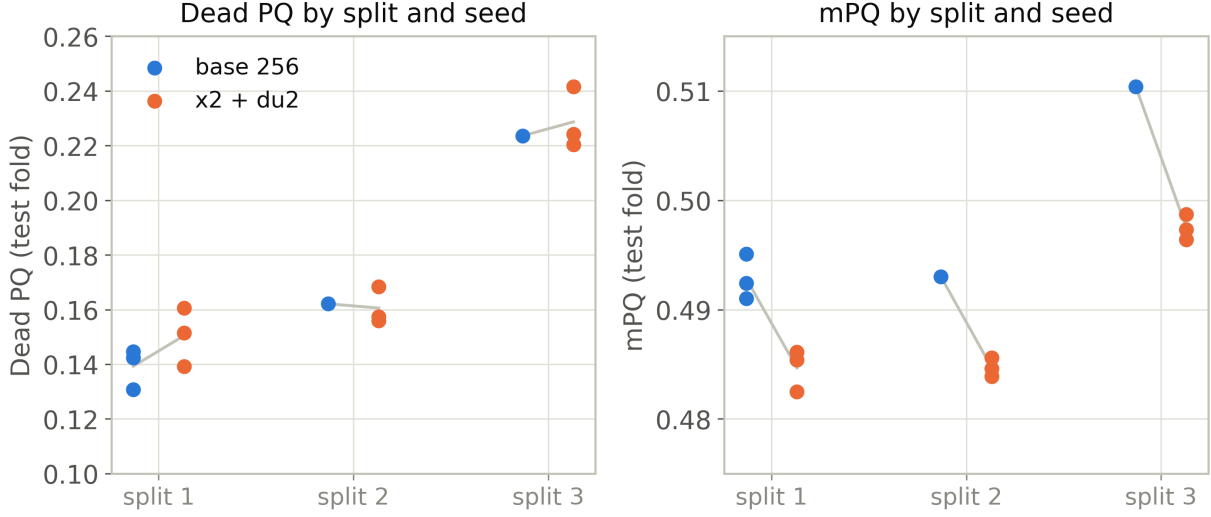


Figure 4: The seed grid. Right: mPQ—every *du2* run (orange) sits below every baseline run (blue) on its split; the price is unambiguous. Left: Dead PQ—runs interleave; split-2’s *du2* seed-mean lands *below* the baseline seed, split-3’s seed 19 flips sign. Lines connect seed means (baseline extra seeds exist on split 1 only).

19 flips, $-.0032$), and split-level seed means go $+.0111$ (split 1, 3/3 seeds) / $-.0015$ (split 2, flips) / $+.0051$ (split 3, 2/3). The x2 Dead advantage is real on split 1, present-but-noisy on split 3, a wash on split 2.

5.6 Final standing: the priced trade

Table 1, bottom block, collects the standing (3-split means, seed 19, no TTA): the levered $2\times$ arm trades $-.0085$ mPQ / $-.0045$ bPQ for interior-Dead miss $-.070$ ($0.372 \rightarrow 0.302$), Dead $F_c +.035$, Dead $DQ^+ +.026$, Uterus-excluded Dead PQ $+.009$, and interior-Dead matched share $+.07$; mPQ $^+$ is at parity ($-.0005$ on the pooled grid; $-.0007$ seed 19). Per the pre-registered combined endpoint, $2\times$ working resolution with the corrected decode is **not a default win**—it is an explicitly priced trade, and the price is now fully accounted for: decode constants (*du2*) $\approx$ two-thirds of the bPQ tax, the lever another eighth, the residual $\approx$ genuine fp-split at 512.

6 Discussion

Why detection-first does not fix Dead. Line A’s two results have one shape. KongNet’s detector finds Dead nuclei (per-nucleus F_c mid-pack) but its decode fuses touching same-class nuclei, and Dead’s failure geometry *is* touching same-class clusters; LKCell-L separates and matches instances better (bPQ, mPQ $^+$, Dead F_c all up) but its detected Dead nuclei fail type pairing, so detection gains do not become Dead PQ. Together with report I’s negative results, the picture is now: the Dead deficit is not a property of one decoder family, and it is not closed by any published architecture under a matched protocol. What *moves* it is resolution—which brings more pixels to bear on small nuclei—and even that helps detection while leaving the separation and typing of what is detected to the same machinery as before.

Table 5: 3×3 seed grid, per-split seed means $\pm$ std (*du2* arm; baseline seeds 1/2 exist on split 1 only). Bottom: 3-split means of seed means, and the *du2lev* arm.

Split	Arm	mPQ	bPQ	Dead PQ
1	base	.4928 $\pm$.0021	.6661 $\pm$.0015	.1393 $\pm$.0074
	du2	.4846 $\pm$.0019	.6565 $\pm$.0003	.1504 $\pm$.0107
2	base	.4930	.6638	.1621
	du2	.4847 $\pm$.0009	.6568 $\pm$.0005	.1606 $\pm$.0068
3	base	.5104	.6647	.2236
	du2	.4975 $\pm$.0012	.6605 $\pm$.0013	.2287 $\pm$.0112
3-split	base	.4987	.6648	.1750
	du2	.4889	.6579	.1799
	du2lev	.4905	.6603	.1801

Pixel-unit constants are a first-class confound. The *du2* result has an implication beyond this project: every pipeline built on HoVer-Net’s post-processing (which is most of the field) carries resolution-hardcoded constants. Any resolution, patch-size, or backbone-stride change silently re-scales the effective decode; published resolution ablations that do not correct for it measure the change *plus* an artefact that points against higher resolution. We found no discussion of this in the nuclei-segmentation literature; the two-line fix is in our released decode.

Honest effect sizes, pre-registered. The seed grid separates what survives pooling from what does not. The prices survive at $\sim 5\times$ seed noise with uniform sign; the Dead PQ gain does not survive at all as a headline (0.4–0.7 $\times$ seed std, one split flipped). What *is* solid on the Dead side are the detection-geometry endpoints—interior-Dead miss $-.070$ and Dead $F_c +.035$, replicated in sign on every run—while the class-PQ aggregation over 65–97 Dead-bearing images remains the noisiest measurement in the benchmark. This is report I’s noise calibration doing its job: the headline number is not the one to trust, and we say which one is.

Evaluation recipes, again. LKCell-L is the second released model (after HoVer-NeXt-T in report I) whose published numbers exceed our strict re-evaluation by more than typical leaderboard gaps (~ 0.015 uniform), and KongNet’s headline Dead advantage does not survive validation-tuned decoding at all. Both gaps are protocol, not pipeline: everything reproduces cleanly once the recipe is matched. The leaderboard’s meaningful resolution is finer than its recipes.

7 Status and Remaining Work

Line A and Line B are complete and closed (A negative; B a priced, fully-replicated trade with the mechanism accounted for). Remaining before the final report: strict re-evaluation of the remaining released per-fold weights (PromptNucSeg, CellVTA), extending the cross-domain rank pool from four architectures to six; the final report’s figure set (tissue $\times$ class heatmaps, error decomposition across architectures); optional additional external sets (MoNuSeg, NuInsSeg); and the phase-2 data direction (larger leak-free training data, pooled external Dead labels) which remains unstarted and is the main open lever for Dead beyond resolution.

References

- [1] J. Gamper, N. Alemi Koohbanani, K. Benet, A. Khuram, N. Rajpoot, “PanNuke: An Open Pan-Cancer Histology Dataset for Nuclei Instance Segmentation and Classification,” *ECDP*, LNCS 11435, pp. 11–19, 2019; and J. Gamper et al., “PanNuke Dataset Extension, Insights and Baselines,” arXiv:2003.10778, 2020.
- [2] S. Graham, Q. D. Vu, S. E. Raza, A. Azam, Y.-W. Tsang, J. Kwak, N. Rajpoot, “HoVer-Net: Simultaneous Segmentation and Classification of Nuclei in Multi-Tissue Histology Images,” *Medical Image Analysis*, 58:101563, 2019.
- [3] F. Hörst, M. Rempe, L. Heine, J. Kleesiek, “CellViT: Vision Transformers for Precise Cell Segmentation and Classification,” *Medical Image Analysis*, 94:103143, 2024.
- [4] F. Hörst, M. Rempe, H. Becker, L. Heine, J. Keyl, J. Kleesiek, “CellViT++: Energy-Efficient and Adaptive Cell Segmentation and Classification Using Foundation Models,” *Computer Methods and Programs in Biomedicine*, 277:109206, 2026.
- [5] E. Baumann, B. Dislich, J. L. Rumberger, J. L. Nagtegaal, M. Rodríguez Martínez, I. Zlobec, “HoVer-NeXt: A Fast Nuclei Segmentation and Classification Pipeline for Next Generation Histopathology,” *MIDL*, PMLR 250:61–86, 2024.
- [6] Z. Shui, Y. Zhang, K. Yao, C. Zhu, S. Zheng, J. Li, H. Li, Y. Sun, R. Guo, L. Yang, “Unleashing the Power of Prompt-Driven Nucleus Instance Segmentation (PromptNucSeg),” *ECCV*, LNCS 15085, pp. 288–304, 2024.
- [7] J. Lv, E. S. Nasir, K. Xu, M. Jahanifar, B. S. Chohan, B. Elhaminia, S. E. A. Raza, “KongNet: A Multi-headed Deep Learning Model for Detection and Classification of Nuclei in Histopathology Images,” arXiv:2510.23559, 2025.
- [8] Z. Cui, J. Yao, L. Zeng, J. Yang, W. Liu, X. Wang, “LKCell: Efficient Cell Nuclei Instance Segmentation with Large Convolution Kernels,” arXiv:2407.18054, 2024.